

TECHNICAL LAB GUIDE

Enterprise Hybrid Infrastructure & Observability Lab

*A Complete Build, Configuration & Validation Guide for Windows Server, Active Directory, Linux,
and Container-Based Observability*

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1. Executive Summary

This document provides complete, professional-grade guidance for building a representative enterprise IT environment inside an isolated virtual lab. It is intended for systems administrators, infrastructure engineers, and students validating hands-on skills across identity management, enterprise networking, Linux server hardening, and modern observability tooling.

The lab integrates three foundational pillars of enterprise infrastructure into a single, cohesive build:

- Centralized Identity and Access Management (IAM) using Microsoft Active Directory Domain Services.
- Linux systems hardening and service delivery using Nginx, UFW, and Docker.
- Enterprise-grade core networking, including centralized DHCP, DNS, and Layer 3/4 segmentation.

OUTCOME

By the end of this guide, you will have a fully functioning domain (corp.local) with a Windows Server 2022 domain controller, a hardened Ubuntu server running a containerized Prometheus/Grafana monitoring stack, and a Windows 10 client that authenticates against the domain and receives enforced Group Policy settings.

2. Core Concepts & Theoretical Background

2.1 Active Directory Domain Services (AD DS)

Active Directory Domain Services is Microsoft's centralized directory service. Rather than managing users, passwords, and permissions independently on individual machines, AD DS stores all network entities — users, groups, computers, printers, and security policies — in a distributed, hierarchical database (ntds.dit). It provides authentication (verifying who you are, via Kerberos or NTLM) and authorization (verifying what you can access) across an entire corporate boundary.

2.2 Domain Controller (DC)

A Domain Controller is the physical or virtual server that actively runs the AD DS role. It acts as the gatekeeper of the corporate forest and domain: it responds to authentication requests, validates credentials, enforces security policy across endpoints, and replicates directory changes with other domain controllers. In this lab, DC-1.corp.local serves as the primary root domain controller.

2.3 Dynamic Host Configuration Protocol (DHCP)

DHCP is a network management protocol (UDP ports 67/68) used to dynamically assign IP addresses and network parameters using the DORA process:

1. Discover — the client broadcasts looking for any available DHCP server.

2. Offer — the DHCP server reserves an IP address and sends an offer packet.
3. Request — the client asks to lease the offered IP address.
4. Acknowledge (ACK) — the DHCP server confirms the reservation and passes network options.

In an enterprise architecture, DHCP also injects critical configuration options:

- Option 003 (Router) — sets the default gateway so machines can reach other subnets or the internet.
- Option 006 (DNS Servers) — informs endpoints to query the Domain Controller for domain-level name resolution.

2.4 Group Policy & Group Policy Objects (GPOs)

Group Policy is a Windows Server feature that enables centralized configuration and management of operating systems, applications, and user settings.

- Group Policy Objects (GPOs) — virtual collections of policy settings, each identified by a unique GUID.
- Inheritance hierarchy (LSDOU) — policies are evaluated and applied in strict order: Local, Site, Domain, and Organizational Unit. Settings applied last (OU-level) override earlier conflicts, ensuring specific business units receive targeted security postures.

3. Architecture & Technology Stack

The table below summarizes every tier of the lab, the operating system or platform used, the installed software, and the network configuration applied to each node.

Tier / Role	OS / Base	Software / Packages	Network Configuration
Hypervisor	Windows 10/11 Host	VMware Workstation Pro	NAT Adapter (VMnet8): 192.168.10.0/24
Domain Controller	Windows Server 2022 Datacenter	AD DS, DNS Server, DHCP Server, GPMC	IP: 192.168.10.10/24 (Static) GW: 192.168.10.2 DNS: 127.0.0.1
Linux Node	Ubuntu Server 22.04 LTS	Nginx, UFW, Docker Engine, Docker Compose, Netplan	IP: 192.168.10.20/24 (Static) GW: 192.168.10.2 DNS: 192.168.10.10
Observability	Containerized (Docker)	Prometheus, Node Exporter, Grafana	Ports: 9090 (Prometheus) 9100 (Node Exporter) 3000 (Grafana)
Workstation	Windows 10 Enterprise	RSOP Client, Edge Browser	Dynamic DHCP (192.168.10.50+) Domain: corp.local

3.1 Logical Network Diagram (Description)

All lab nodes reside on a single isolated NAT segment (192.168.10.0/24) provided by VMware's VMnet8. The Domain Controller (192.168.10.10) is authoritative for DNS and DHCP; the Ubuntu Linux node (192.168.10.20) holds a static address and consumes DNS from the Domain Controller; the Windows 10 workstation receives a dynamic lease in the 192.168.10.50–150 range and resolves the corp.local domain through the same Domain Controller. All outbound internet traffic routes through the shared NAT gateway at 192.168.10.2.

4. Prerequisites & Preparation

Before beginning implementation, confirm the following are available:

- A Windows 10/11 host machine with virtualization (Intel VT-x / AMD-V) enabled in firmware.
- VMware Workstation Pro installed and licensed.
- ISO media for Windows Server 2022 Datacenter, Windows 10 Enterprise, and Ubuntu Server 22.04 LTS.
- A minimum of 16 GB host RAM and 150 GB free disk space recommended for running three concurrent VMs comfortably.
- Administrative rights on the host machine to configure VMware's virtual network editor.

IMPORTANT

The Domain Controller must be the sole authority for DHCP on this network segment. Running VMware's built-in NAT DHCP service alongside the Windows DHCP server will cause duplicate IP address assignment and intermittent client connectivity failures.

5. Implementation Guide

Phase 1 — Isolated Network Configuration

This phase configures VMware's virtual routing so that all lab virtual machines are isolated on a private segment while still retaining outbound internet access.

1. Open VMware Workstation, then navigate to Edit → Virtual Network Editor → Change Settings.
2. Select VMnet8 (NAT) and set the Subnet IP to 192.168.10.0 with Subnet Mask 255.255.255.0.
3. Click NAT Settings..., set the Gateway IP to 192.168.10.2, and click OK.
4. Uncheck "Use local DHCP service to distribute IP addresses to VMs." The Windows Domain Controller must hold sole authority over DHCP distribution to prevent race conditions.
5. Click Apply, then OK to save the configuration.

Phase 2 — Building the Windows Server 2022 Domain Controller (DC-1)

Step 1: Assign the Static IP Address

1. Open the Network Connections applet by running `ncpa.cpl`.
2. Right-click the Ethernet adapter, select Properties, then select Internet Protocol Version 4 (TCP/IPv4) → Properties.
3. Configure the static parameters as follows:
 - IP Address: 192.168.10.10
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.10.2
 - Preferred DNS Server: 127.0.0.1 (points the server to its own internal DNS service once installed)

Step 2: Role Installation

1. Launch Server Manager → Manage → Add Roles and Features.
2. Advance to Server Roles and select Active Directory Domain Services and DHCP Server. DNS Server installs automatically as an AD DS dependency.
3. Accept all required features and complete the installation wizard.

Step 3: Domain Forest Promotion

1. In Server Manager, click the Deployment Notification flag and select Promote this server to a domain controller.
2. Select Add a new forest and enter the root domain name: corp.local.
3. Set a Directory Services Restore Mode (DSRM) recovery password.
4. Advance through the default options; the NetBIOS domain name auto-assigns as CORP.
5. Click Install. The server reboots automatically and initializes the directory schema.

Step 4: Enterprise DHCP Scope Creation & Authorization

1. Open DHCP Management by running `dhcpcmgmt.msc`.
2. Right-click IPv4 and select New Scope...
3. Set the Scope Name to Corp_LAN.
4. Configure the address pool distribution:
 - Start IP: 192.168.10.50
 - End IP: 192.168.10.150
 - Subnet Mask: 255.255.255.0 (Length: 24)
5. Under DHCP Options, set Option 003 (Router) to 192.168.10.2 and confirm Option 006 (DNS) lists 192.168.10.10 as the primary server.
6. Activate the scope.

7. Right-click dc-1.corp.local in the DHCP tree and select Authorize.

Step 5: Directory Structure & User Provisioning

1. Open Active Directory Users and Computers by running dsa.msc.
2. Right-click the root domain corp.local, select New → Organizational Unit, and name it Corp_Employees.
3. Right-click Corp_Employees, select New → Organizational Unit, and name it IT_Dept.
4. Inside IT_Dept, right-click and select New → User, then configure:
 - Full Name: Alex Morgan
 - User logon name: amorgan
 - Initial Password: set to meet domain complexity requirements (e.g., P@ssword2026!)
 - Check “User must change password at next logon”

Step 6: Security Group Policy Configuration

1. Open Group Policy Management by running gpmc.msc.
2. Expand Domains → corp.local.
3. Right-click the Corp_Employees OU and select Create a GPO in this domain, and Link it here...
4. Name the GPO GPO_Workstation_Security_Baseline.
5. Right-click the GPO and select Edit.
6. Navigate to User Configuration → Policies → Administrative Templates → Control Panel → Personalization and configure:
 - Enable screen saver: Enabled
 - Password protect the screen saver: Enabled
 - Screen saver timeout: Enabled, set to 900 seconds (15-minute compliance enforcement)

Phase 3 — Linux Server Hardening & Observability (Ubuntu 22.04 LTS)

Step 1: Static IP Configuration via Netplan

1. Edit the Netplan interface configuration file:

```
sudo nano /etc/netplan/00-installer-config.yaml
```

2. Populate the interface definition (replace ens33 with your primary adapter name):

```
network:
  version: 2
  ethernets:
    ens33:
      dhcp4: no
      addresses:
        - 192.168.10.20/24
      routes:
        - to: default
          via: 192.168.10.2
      nameservers:
        addresses: [192.168.10.10, 8.8.8.8]
```

3. Apply the new network state:

```
sudo netplan apply
```

Step 2: Nginx Web Server & Firewall Isolation (UFW)

1. Install and enable the web server:

```
sudo apt update && sudo apt install nginx curl -y
sudo systemctl enable --now nginx
```

2. Enforce strict perimeter firewall rules:

```
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw allow 22/tcp comment 'SSH'
sudo ufw allow 80/tcp comment 'Nginx HTTP'
sudo ufw allow 3000/tcp comment 'Grafana UI'
sudo ufw enable
```

Step 3: Containerized Monitoring Deployment (Docker Compose)

1. Install the container runtime and add your user to the docker group:

```
sudo apt install docker.io docker-compose-v2 -y
sudo usermod -aG docker $USER
newgrp docker
```

2. Create the monitoring workspace:

```
mkdir -p ~/monitoring && cd ~/monitoring
```

3. Create the Prometheus scrape configuration (prometheus.yml):

```
global:
  scrape_interval: 15s

scrape_configs:
  - job_name: 'node_exporter'
    static_configs:
      - targets: ['node_exporter:9100']
```

4. Define the multi-container stack (docker-compose.yml):

```
services:
  prometheus:
    image: prom/prometheus:latest
    container_name: prometheus
    restart: unless-stopped
    volumes:
      - ./prometheus.yml:/etc/prometheus/prometheus.yml
    ports:
      - "9090:9090"

  node_exporter:
    image: prom/node-exporter:latest
    container_name: node_exporter
    restart: unless-stopped
    ports:
      - "9100:9100"

  grafana:
    image: grafana/grafana:latest
    container_name: grafana
    restart: unless-stopped
    ports:
      - "3000:3000"
    environment:
      - GF_SECURITY_ADMIN_USER=admin
      - GF_SECURITY_ADMIN_PASSWORD=admin
```

5. Deploy the stack in detached mode:

```
docker compose up -d
```

Phase 4 — Client Join, Verification & Proof-of-Work (Windows 10)

Step 1: Dynamic Addressing & Domain Join

1. Set the Windows 10 network interface to obtain an IP address automatically via DHCP.
2. Confirm dynamic lease allocation with `ipconfig /all` — the client should receive an address between 192.168.10.50 and .150, with DNS pointing to 192.168.10.10.
3. Open System Properties (`sysdm.cpl`) and click Change...
4. Select Domain, enter `corp.local`, and supply `CORP\Administrator` credentials.
5. Reboot the workstation upon seeing the domain welcome notification.

Step 2: User Authentication & Group Policy Verification

1. At the Windows sign-in screen, select Other user.
2. Log in using `amorgan` and the initial password.
3. Complete the forced password change prompt to satisfy the domain account policy.
4. Open Command Prompt and verify the active identity:

```
whoami
```

Expected output: corp\amorgan

5. Generate the Resultant Set of Policy report:

```
gpresult /r
```

VERIFICATION CRITERIA

Under USER SETTINGS, the section Applied Group Policy Objects must display GPO_Workstation_Security_Baseline, confirming policy application from `dc-1.corp.local`.

Step 3: Observability Validation Across Subnets

1. Launch Microsoft Edge on the Windows 10 client workstation.
2. Navigate to `http://192.168.10.20` to confirm reachability through Nginx and the UFW firewall.
3. Navigate to `http://192.168.10.20:3000` to confirm direct access to Grafana.
4. Log in with `admin / admin`, configure the Prometheus datasource (`http://prometheus:9090`), and import Dashboard ID 1860 to view real-time CPU, RAM, and disk telemetry from the Linux host.

6. Final Verification Checklist

Use this checklist to confirm every component of the build is functioning as designed before considering the lab complete.

#	Verification Item	Expected Result
1	VMnet8 network configuration	Subnet 192.168.10.0/24, local DHCP disabled on host
2	Domain Controller static IP	192.168.10.10, self-referencing DNS (127.0.0.1)
3	AD DS forest promotion	Forest corp.local created and operational
4	DHCP scope status	Corp_LAN scope active and authorized
5	OU structure	Corp_Employees → IT_Dept created with user amorgan
6	GPO linkage	GPO_Workstation_Security_Baseline linked to Corp_Employees
7	Linux static IP	192.168.10.20, DNS pointing to domain controller
8	Firewall rules (UFW)	Only ports 22, 80, and 3000 permitted inbound
9	Docker stack	prometheus, node_exporter, and grafana containers running
10	Client domain join	whoami returns corp\amorgan
11	Group Policy application	gpresult /r lists the baseline GPO
12	Cross-subnet reachability	Nginx and Grafana both reachable from the Windows client

7. Appendix — Quick Reference

7.1 IP Address Summary

Node	IP Address	Role
VMware Gateway (VMnet8)	192.168.10.2	NAT gateway for all lab traffic
Domain Controller (DC-1)	192.168.10.10	AD DS, DNS, DHCP
Linux Node	192.168.10.20	Nginx, Docker, Observability stack
Windows 10 Client	192.168.10.50 – 150 (dynamic)	Domain-joined workstation

7.2 Service Port Reference

Service	Port	Protocol
SSH	22	TCP
HTTP (Nginx)	80	TCP
Prometheus	9090	TCP
Node Exporter	9100	TCP
Grafana	3000	TCP

7.3 Key Credentials Used in This Lab

For lab purposes only — replace with strong, unique credentials in any non-lab environment.

Account	Username	Notes
Domain Administrator	CORP\Administrator	Used for domain join operations
Standard Domain User	amorgan	Forced password change at first logon
Grafana Admin	admin / admin	Change immediately after first login